NETWORK SECURITY ANALYSIS REPORT

test_flows.csv

Analysis #1 (2026-10-26T15:26:29 UTC) · report #1 · generated 2026-10-26T15:27:04+00:00 UTC · schema 2.0 · detector LightGBM v3 (class offsets, chosen without test labels)

What did the detector predict, what evidence supports it, what remains uncertain, and what should an analyst check next?

A. Summary

Offline analysis of 300 completed network flows uploaded as "test_flows.csv". The file contains flow measurements only: no live traffic, IP addresses, ports, timestamps or payloads.

Predicted Normal	Predicted attack	Needs review (confidence < 70%)	Mean confidence
34 (11.3%)	**266** (88.7%)	**151**	**73.0%**

Batch triage priority: CRITICAL (rule: serious flows >= 25% of all flows, or malicious flows >= 50%)

Batch triage priority: an application heuristic that orders review work from the predicted classes and fixed severity rules. It is not a model output, not observed business impact and not a confirmed compromise.

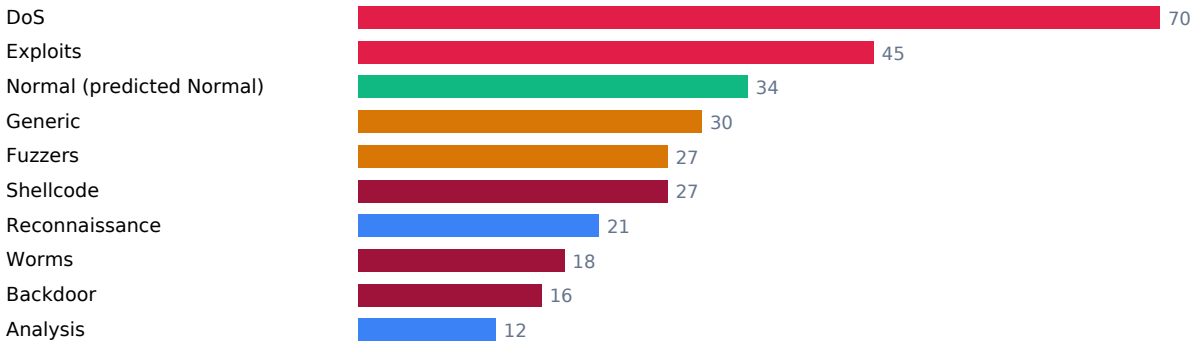
Most important uncertainty. 151 predictions have confidence below 70%.

Next check. Check flow #79 (tcp/-, state FIN) against packet captures, host logs or sensor alerts from the same capture period to confirm whether it is DoS activity.

Component status

Component	Status	Detail
Detection	available	300 flows analysed in 0.1 s
Explanations (XAI)	available	5 of 5 selected flows explained (TreeSHAP); LIME not generated
Retrieval (RAG)	available	23 distinct passages from 5 source families
LLM generation	available	generated: 4
Automated checks	available	22 generated statements checked, 1 rejected
PDF rendering	available	

B. Detections and uncertainty



Predicted class	Flows	Share	Mean conf.	Severity (heuristic)	Status uncertain	Type uncertain
DoS	70	23.3%	56.1%	High	0	69
Exploits	45	15.0%	66.5%	High	0	25
Normal	34	11.3%	87.3%	-	7	0
Generic	30	10.0%	98.4%	Medium	0	0
Fuzzers	27	9.0%	71.4%	Medium	0	5

Predicted class	Flows	Share	Mean conf.	Severity (heuristic)	Status uncertain	Type uncertain
Shellcode	27	9.0%	71.8%	Critical	0	6
Reconnaissance	21	7.0%	97.1%	Low	0	1
Worms	18	6.0%	77.7%	Critical	0	7
Backdoor	16	5.3%	65.6%	Critical	0	8
Analysis	12	4.0%	58.8%	Low	0	12

Prediction sets (90% shortlists)

Shortlist	Flows
Single class	160
Several classes	140
Contains Normal	32
Attack STATUS uncertain (Normal and an attack type)	7
Attack TYPE uncertain (several attack types, no Normal)	133

• Probabilities are calibrated on held-out rows of the training file (temperature scaling). They describe the model, not proof, and can be less reliable on traffic that differs from the training data.

• The 90% prediction set (shortlist) targets 90% coverage on average over flows like the calibration rows; it is not a per-flow guarantee, and a single-class set is not certainty.

• Attack-STATUS uncertain: the shortlist contains Normal and at least one attack type (attack or not is unclear). Attack-TYPE uncertain: predicted attack and the shortlist has several attack types but not Normal.

• Confidence, heuristic severity and business impact are different quantities.

• On the official test file (shifted from the training file) the detector flagged 27.2% of Normal flows as attacks (model card).

• For the previous detector v2 the 90% sets contained the true class for 79.1% of official test flows (Notebook 21); this was not re-measured for v3.

Review order: flows to look at first

Flow	Predicted class	Severity (heuristic)	Conf.	Shortlist	proto / service	Bytes src / dst
#172	Worms	Critical	99.4%	Worms	tcp / http	2,416 / 308,476
#18	Backdoor	Critical	98.7%	Backdoor	srp / -	200 / 0
#283	Worms	Critical	98.6%	Worms	tcp / http	3,114 / 482,584
#178	Worms	Critical	98.5%	Worms	tcp / http	2,314 / 308,456
#112	Worms	Critical	98.5%	Worms	tcp / http	1,302 / 268
#87	Backdoor	Critical	98.3%	Backdoor	etherip / -	200 / 0
#168	Worms	Critical	97.3%	Worms	tcp / http	1,282 / 268
#6	Worms	Critical	91.8%	Worms	tcp / http	1,296 / 268
#77	Backdoor	Critical	91.8%	Backdoor	fire / -	200 / 0
#64	Shellcode	Critical	91.7%	Shellcode	udp / -	196 / 0

Ordered by heuristic severity, then confidence. This is a review order, not a list of confirmed threats.

C. Why the model predicted this

Each card is **one selected flow**; the selection rule is stated on the card. Individual-flow evidence is kept separate from the batch summary. Contributions explain the detection model, not the cause of the traffic.

Flow #79: predicted DoS (calibrated probability 0.79)

Why this flow: illustrative flow for DoS: the DoS flow with the highest confidence (1 of 70 flows of this type; it does not represent every one).

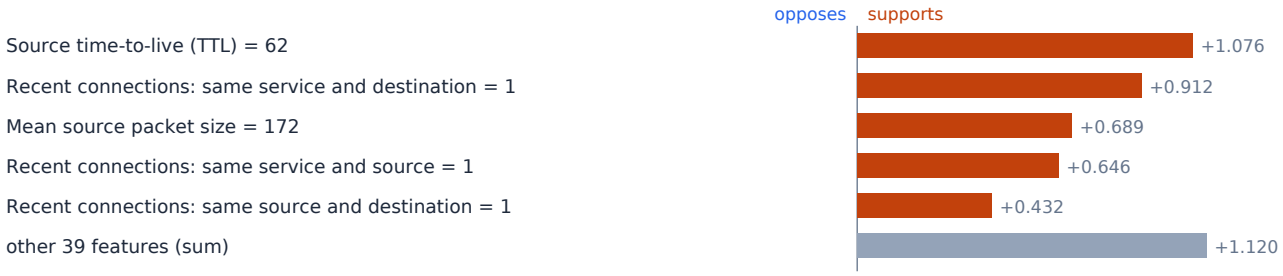
Class	Probability	On 90% shortlist
DoS	79.4%	yes
Exploits	18.6%	no
Normal	1.2%	no
Generic	0.4%	no

Shortlist: DoS · Normal on shortlist: no · confidence level medium · heuristic severity High

Measured values (observed in the upload)

Feature	Value	Compared with training flows
Transport protocol proto	tcp	45.6% of training flows have this value
Application service service	-	53.6% of training flows have this value
Connection state state	FIN	44.4% of training flows have this value
Flow duration dur	0.4676	higher than about 70% of training flows
Bytes from source sbytes	1724	higher than about 80% of training flows
Bytes from destination dbytes	268	between the 54th and 60th percentile of training flows (a very common value)
Packets from source spkts	10	between the 56th and 72nd percentile of training flows (a very common value)
Packets from destination dpkts	6	between the 56th and 63rd percentile of training flows (a very common value)
Source time-to-live (TTL) sttl	62	between the 25th and 35th percentile of training flows (a very common value)
Destination time-to-live (TTL) dttl	252	between the 72nd and 100th percentile of training flows (a very common value)
Mean source packet size smean	172	higher than about 89% of training flows
Mean destination packet size dmean	45	between the 56th and 63rd percentile of training flows (a very common value)

Model attribution (TreeSHAP)



Base value -4.760 + all contributions = model score +0.116; deployed class score = +0.116 / 1.159 + offset +0.80 = +0.900. All shown contributions have the same sign; no opposing evidence among the top features. Each contribution moves the model score of the predicted class up (supports) or down (opposes). Base value + all contributions = the model score; the deployed detector divides it by the temperature and adds a fixed class offset before turning scores into probabilities. Contributions are not percentages of the probability, and they describe the model, not the cause of the traffic.

Feature	Value	Contrib.	Direction	Meaning
Source time-to-live (TTL)	62	+1.076	supports	Source to destination time to live value
Recent connections: same service and destination	1	+0.912	supports	No. of connections that contain the same service (14) and destination address (3) in 100 connections according to the last time (26).
Mean source packet size	172	+0.689	supports	Mean of the flow packet size transmitted by the src
Recent connections: same service and source	1	+0.646	supports	No. of connections that contain the same service (14) and source address (1) in 100 connections according to the last time (26).
Recent connections: same source and destination	1	+0.432	supports	No of connections of the same source (1) and the destination (3) address in in 100 connections according to the last time (26).

LIME: not generated — LIME was only run in research Notebook 16 for an older detector; it is not computed per flow for the deployed model.

Evidence given to the LLM

Evidence id	Fact (computed by code)
F79.E1	Predicted class: DoS with calibrated probability 0.79
F79.E2	Conformal prediction set (90% coverage): DoS
F79.E3	Column sttl = 62 supports DoS (contribution +1.076)
F79.E4	Column ct_srv_dst = 1 supports DoS (contribution +0.912)
F79.E5	Column smean = 172 supports DoS (contribution +0.689)
F79.E6	Column ct_srv_src = 1 supports DoS (contribution +0.646)
F79.E7	Column ct_dst_src_ltm = 1 supports DoS (contribution +0.432)
F79.E8	In the analysed dataset, 70 flows (26.3% of the predicted attacks) were classified as DoS, with average confidence 0.56

Plain-language explanation

The model is 0.79 confident that this flow is DoS with the 90% prediction set containing DoS. The strongest evidence is sttl = 62 (contribution +1.076) and ct_srv_dst = 1 (contribution +0.912), while smean = 172 (contribution +0.689) and ct_srv_src = 1 (contribution +0.646) also support it. In the analysed dataset, 70 flows (26.3% of the predicted attacks) were classified as DoS, with average confidence 0.56. [F79.E1] [F79.E2] [F79.E3] [F79.E4] [F79.E5] [F79.E6] [F79.E8] (LLM, passed automated checks; supported; citations completed by the verifier (E6))

Generation: status generated, model qwen2.5:7b, 13 s, 8 statements checked, 1 rejected.

Rejected (techniques, round 1): not a real ATT&CK ID — “S1”

Flow #222: predicted Exploits (calibrated probability 0.99)

Why this flow: illustrative flow for Exploits: the Exploits flow with the highest confidence (1 of 45 flows of this type; it does not represent every one).

Class	Probability	On 90% shortlist
Exploits	98.6%	yes
DoS	1.1%	no
Normal	0.2%	no
Generic	0.0%	no

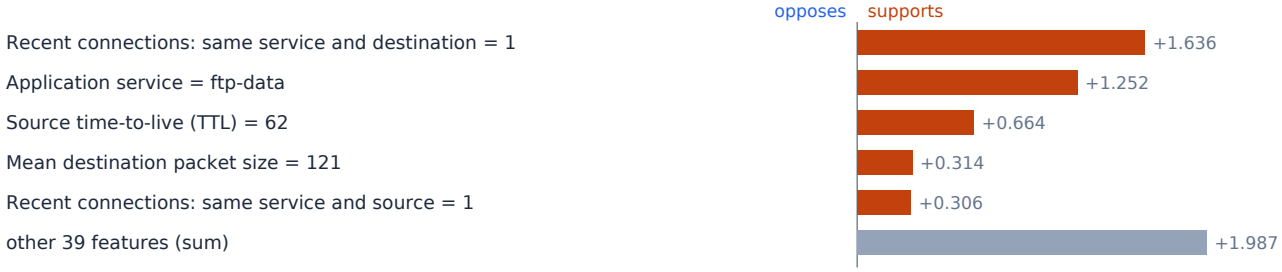
Shortlist: Exploits · Normal on shortlist: no · confidence level high · heuristic severity High

Measured values (observed in the upload)

Feature	Value	Compared with training flows
Transport protocol proto	tcp	45.6% of training flows have this value
Application service service	ftp-data	2.3% of training flows have this value

Feature	Value	Compared with training flows
Connection state state	FIN	44.4% of training flows have this value
Flow duration dur	0.7952	higher than about 79% of training flows
Bytes from source sbytes	364	higher than about 49% of training flows
Bytes from destination dbytes	968	higher than about 75% of training flows
Packets from source spkts	8	higher than about 55% of training flows
Packets from destination dpkts	8	between the 63rd and 73rd percentile of training flows (a very common value)
Source time-to-live (TTL) sttl	62	between the 25th and 35th percentile of training flows (a very common value)
Destination time-to-live (TTL) dttl	252	between the 72nd and 100th percentile of training flows (a very common value)
Mean source packet size smean	46	higher than about 2% of training flows
Mean destination packet size dmean	121	higher than about 83% of training flows

Model attribution (TreeSHAP)



Base value -3.721 + all contributions = model score +2.437; deployed class score = +2.437 / 1.159 + offset +0.00 = +2.103. All shown contributions have the same sign; no opposing evidence among the top features. Each contribution moves the model score of the predicted class up (supports) or down (opposes). Base value + all contributions = the model score; the deployed detector divides it by the temperature and adds a fixed class offset before turning scores into probabilities. Contributions are not percentages of the probability, and they describe the model, not the cause of the traffic.

Feature	Value	Contrib.	Direction	Meaning
Recent connections: same service and destination	1	+1.636	supports	No. of connections that contain the same service (14) and destination address (3) in 100 connections according to the last time (26).
Application service	ftp-data	+1.252	supports	http, ftp, smtp, ssh, dns, ftp-data ,irc and (-) if not much used service
Source time-to-live (TTL)	62	+0.664	supports	Source to destination time to live value
Mean destination packet size	121	+0.314	supports	Mean of the flow packet size transmitted by the dst
Recent connections: same service and source	1	+0.306	supports	No. of connections that contain the same service (14) and source address (1) in 100 connections according to the last time (26).

LIME: not generated — LIME was only run in research Notebook 16 for an older detector; it is not computed per flow for the deployed model.

Evidence given to the LLM

Evidence id	Fact (computed by code)
F222.E1	Predicted class: Exploits with calibrated probability 0.99
F222.E2	Conformal prediction set (90% coverage): Exploits

Evidence id	Fact (computed by code)
F222.E3	Column ct_srv_dst = 1 supports Exploits (contribution +1.636)
F222.E4	Column service = ftp-data supports Exploits (contribution +1.252)
F222.E5	Column sttl = 62 supports Exploits (contribution +0.664)
F222.E6	Column dmean = 121 supports Exploits (contribution +0.314)
F222.E7	Column ct_srv_src = 1 supports Exploits (contribution +0.306)
F222.E8	In the analysed dataset, 45 flows (16.9% of the predicted attacks) were classified as Exploits, with average confidence 0.67

Plain-language explanation

The model is 0.99 confident that this flow is Exploits with the 90% prediction set containing Exploits. The strongest evidence is ct_srv_dst = 1 (contribution +1.636) and service = ftp-data (contribution +1.252), while sttl = 62 (contribution +0.664) supports it. In the analysed dataset, 45 flows (16.9% of the predicted attacks) were classified as Exploits, with average confidence 0.67. [F222.E1] [F222.E2] [F222.E3] [F222.E4] [F222.E5] [F222.E8] (LLM, passed automated checks; supported)

Generation: status generated, model qwen2.5:7b, 7 s, 5 statements checked, 0 rejected.

Flow #142: predicted Generic (calibrated probability 1.00)

Why this flow: illustrative flow for Generic: the Generic flow with the highest confidence (1 of 30 flows of this type; it does not represent every one).

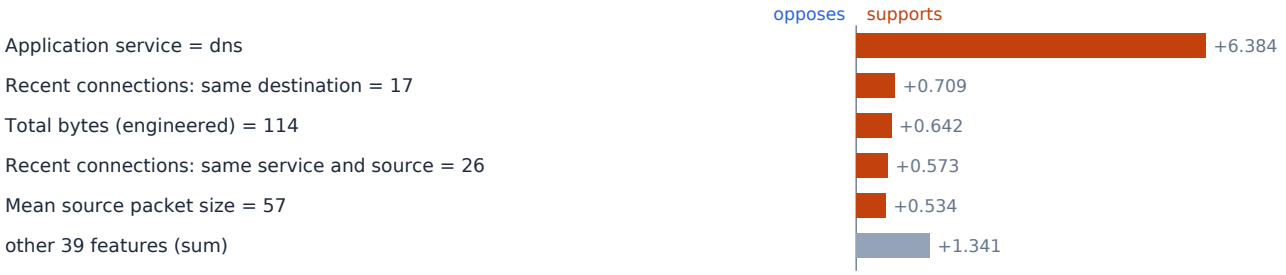
Class	Probability	On 90% shortlist
Generic	100.0%	yes
Fuzzers	0.0%	no
Exploits	0.0%	no
DoS	0.0%	no

Shortlist: Generic · Normal on shortlist: no · confidence level high · heuristic severity Medium

Measured values (observed in the upload)

Feature	Value	Compared with training flows
Transport protocol proto	udp	36.1% of training flows have this value
Application service service	dns	27.0% of training flows have this value
Connection state state	INT	46.8% of training flows have this value
Flow duration dur	0.000008	between the 23rd and 29th percentile of training flows (a very common value)
Bytes from source sbytes	114	between the 5th and 27th percentile of training flows (a very common value)
Bytes from destination dbytes	0	between the 0th and 48th percentile of training flows (a very common value)
Packets from source spkts	2	between the 2nd and 51st percentile of training flows (a very common value)
Packets from destination dpkts	0	between the 0th and 48th percentile of training flows (a very common value)
Source time-to-live (TTL) sttl	254	between the 35th and 100th percentile of training flows (a very common value)
Destination time-to-live (TTL) dttl	0	between the 0th and 48th percentile of training flows (a very common value)
Mean source packet size smean	57	between the 15th and 38th percentile of training flows (a very common value)
Mean destination packet size dmean	0	between the 0th and 48th percentile of training flows (a very common value)

Model attribution (TreeSHAP)



Base value -3.962 + all contributions = model score +6.222; deployed class score = +6.222 / 1.159 + offset +0.20 = +5.568. All shown contributions have the same sign; no opposing evidence among the top features. Each contribution moves the model score of the predicted class up (supports) or down (opposes). Base value + all contributions = the model score; the deployed detector divides it by the temperature and adds a fixed class offset before turning scores into probabilities. Contributions are not percentages of the probability, and they describe the model, not the cause of the traffic.

Feature	Value	Contrib.	Direction	Meaning
Application service	dns	+6.384	supports	http, ftp, smtp, ssh, dns, ftp-data ,irc and (-) if not much used service
Recent connections: same destination	17	+0.709	supports	No. of connections of the same destination address (3) in 100 connections according to the last time (26).
Total bytes (engineered)	114	+0.642	supports	Engineered feature: bytes from source + bytes from destination (sbytes + dbytes).
Recent connections: same service and source	26	+0.573	supports	No. of connections that contain the same service (14) and source address (1) in 100 connections according to the last time (26).
Mean source packet size	57	+0.534	supports	Mean of the flow packet size transmitted by the src

LIME: not generated — LIME was only run in research Notebook 16 for an older detector; it is not computed per flow for the deployed model.

Evidence given to the LLM

Evidence id	Fact (computed by code)
F142.E1	Predicted class: Generic with calibrated probability 1.00
F142.E2	Conformal prediction set (90% coverage): Generic
F142.E3	Column service = dns supports Generic (contribution +6.384)
F142.E4	Column ct_dst_ltm = 17 supports Generic (contribution +0.709)
F142.E5	Column fe_total_bytes = 114 supports Generic (contribution +0.642)
F142.E6	Column ct_srv_src = 26 supports Generic (contribution +0.573)
F142.E7	Column smean = 57 supports Generic (contribution +0.534)
F142.E8	In the analysed dataset, 30 flows (11.3% of the predicted attacks) were classified as Generic, with average confidence 0.98

Plain-language explanation

The model is 1.00 confident that this flow is Generic, and its 90% prediction set contains Generic. The strongest evidence is service = dns (contribution +6.384) and ct_dst_ltm = 17 (contribution +0.709), while fe_total_bytes = 114 (contribution +0.642) supports it. In the analysed dataset, 30 flows (11.3% of the predicted attacks) were classified as Generic, with average confidence 0.98. [F142.E1] [F142.E2] [F142.E3] [F142.E4] [F142.E5] [F142.E8] (LLM, passed automated checks; supported)

Generation: status generated, model qwen2.5:7b, 7 s, 5 statements checked, 0 rejected.

Flow #161: predicted Fuzzers (calibrated probability 0.96)

Why this flow: illustrative flow for Fuzzers: the Fuzzers flow with the highest confidence (1 of 27 flows of this type; it does not represent every one).

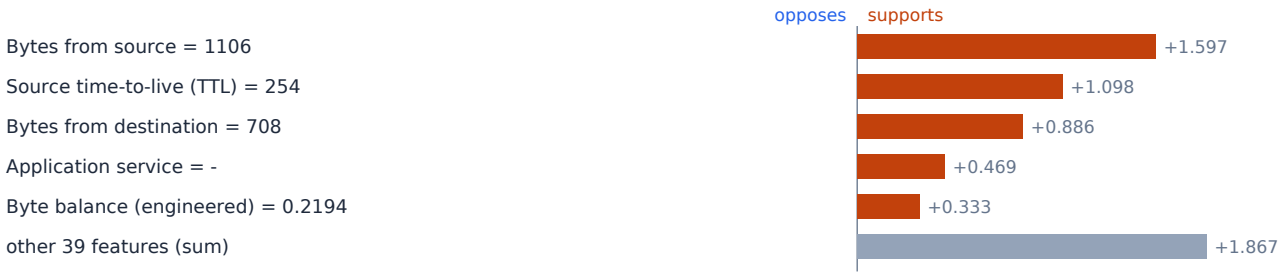
Class	Probability	On 90% shortlist
Fuzzers	96.0%	yes
Normal	3.3%	no
Exploits	0.4%	no
DoS	0.2%	no

Shortlist: Fuzzers · Normal on shortlist: no · confidence level high · heuristic severity Medium

Measured values (observed in the upload)

Feature	Value	Compared with training flows
Transport protocol proto	tcp	45.6% of training flows have this value
Application service service	-	53.6% of training flows have this value
Connection state state	FIN	44.4% of training flows have this value
Flow duration dur	0.7922	higher than about 79% of training flows
Bytes from source sbytes	1106	higher than about 72% of training flows
Bytes from destination dbytes	708	higher than about 72% of training flows
Packets from source spkts	12	between the 72nd and 76th percentile of training flows (a very common value)
Packets from destination dpkts	8	between the 63rd and 73rd percentile of training flows (a very common value)
Source time-to-live (TTL) sttl	254	between the 35th and 100th percentile of training flows (a very common value)
Destination time-to-live (TTL) dttl	252	between the 72nd and 100th percentile of training flows (a very common value)
Mean source packet size smean	92	higher than about 63% of training flows
Mean destination packet size dmean	89	higher than about 75% of training flows

Model attribution (TreeSHAP)



Base value -4.651 + all contributions = model score +1.599; deployed class score = +1.599 / 1.159 + offset -0.20 = +1.179. All shown contributions have the same sign; no opposing evidence among the top features. Each contribution moves the model score of the predicted class up (supports) or down (opposes). Base value + all contributions = the model score; the deployed detector divides it by the temperature and adds a fixed class offset before turning scores into probabilities. Contributions are not percentages of the probability, and they describe the model, not the cause of the traffic.

Feature	Value	Contrib.	Direction	Meaning
Bytes from source	1106	+1.597	supports	Source to destination transaction bytes
Source time-to-live (TTL)	254	+1.098	supports	Source to destination time to live value

Feature	Value	Contrib.	Direction	Meaning
Bytes from destination	708	+0.886	supports	Destination to source transaction bytes
Application service	-	+0.469	supports	http, ftp, smtp, ssh, dns, ftp-data ,irc and (-) if not much used service
Byte balance (engineered)	0.2194	+0.333	supports	Engineered feature: (sbytes - dbytes) / (sbytes + dbytes), from -1 (all bytes from the destination) to 1 (all from the source); 0 when no bytes were sent.

LIME: not generated — LIME was only run in research Notebook 16 for an older detector; it is not computed per flow for the deployed model.

Evidence given to the LLM

Evidence id	Fact (computed by code)
F161.E1	Predicted class: Fuzzers with calibrated probability 0.96
F161.E2	Conformal prediction set (90% coverage): Fuzzers
F161.E3	Column sbytes = 1106 supports Fuzzers (contribution +1.597)
F161.E4	Column sttl = 254 supports Fuzzers (contribution +1.098)
F161.E5	Column dbytes = 708 supports Fuzzers (contribution +0.886)
F161.E6	Column service = - supports Fuzzers (contribution +0.469)
F161.E7	Column fe_byte_balance = 0.2194 supports Fuzzers (contribution +0.333)
F161.E8	In the analysed dataset, 27 flows (10.2% of the predicted attacks) were classified as Fuzzers, with average confidence 0.71

Plain-language explanation

The model is 0.96 confident that this flow is Fuzzers, and its 90% prediction set contains Fuzzers. The strongest evidence is sbytes = 1106 (contribution +1.597) and sttl = 254 (contribution +1.098), while dbytes = 708 (contribution +0.886) and service = - (contribution +0.469) support Fuzzers. In the analysed dataset, 27 flows (10.2% of the predicted attacks) were classified as Fuzzers, with average confidence 0.71. [F161.E1] [F161.E2] [F161.E3] [F161.E4] [F161.E5] [F161.E6] [F161.E7] [F161.E8] (LLM, passed automated checks; supported)

Generation: status generated, model qwen2.5:7b, 7 s, 4 statements checked, 0 rejected.

Flow #241: predicted Normal (calibrated probability 0.53)

Why this flow: uncertain flow: the lowest-confidence flow whose 90% shortlist contains both Normal and an attack type (attack status uncertain).

Class	Probability	On 90% shortlist
Normal	53.3%	yes
Fuzzers	37.9%	yes
DoS	3.3%	no
Shellcode	3.1%	no

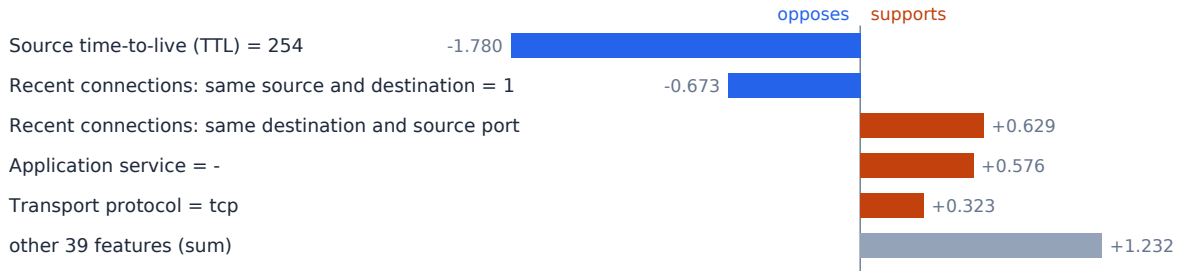
Shortlist: Fuzzers, Normal · Normal on shortlist: yes · confidence level low · heuristic severity None

Measured values (observed in the upload)

Feature	Value	Compared with training flows
Transport protocol proto	tcp	45.6% of training flows have this value
Application service service	-	53.6% of training flows have this value
Connection state state	FIN	44.4% of training flows have this value
Flow duration dur	1.0064	higher than about 83% of training flows
Bytes from source sbytes	534	higher than about 52% of training flows
Bytes from destination dbytes	268	between the 54th and 60th percentile of training flows (a very common value)

Feature	Value	Compared with training flows
Packets from source spkts	10	between the 56th and 72nd percentile of training flows (a very common value)
Packets from destination dpkts	6	between the 56th and 63rd percentile of training flows (a very common value)
Source time-to-live (TTL) sttl	254	between the 35th and 100th percentile of training flows (a very common value)
Destination time-to-live (TTL) dttl	252	between the 72nd and 100th percentile of training flows (a very common value)
Mean source packet size smean	53	higher than about 8% of training flows
Mean destination packet size dmean	45	between the 56th and 63rd percentile of training flows (a very common value)

Model attribution (TreeSHAP)



Base value -2.410 + all contributions = model score -2.103; deployed class score = -2.103 / 1.159 + offset +0.00 = -1.814. Each contribution moves the model score of the predicted class up (supports) or down (opposes). Base value + all contributions = the model score; the deployed detector divides it by the temperature and adds a fixed class offset before turning scores into probabilities. Contributions are not percentages of the probability, and they describe the model, not the cause of the traffic.

Feature	Value	Contrib.	Direction	Meaning
Source time-to-live (TTL)	254	-1.780	opposes	Source to destination time to live value
Recent connections: same source and destination	1	-0.673	opposes	No of connections of the same source (1) and the destination (3) address in in 100 connections according to the last time (26).
Recent connections: same destination and source port	1	+0.629	supports	No of connections of the same destination address (3) and the source port (2) in 100 connections according to the last time (26).
Application service	-	+0.576	supports	http, ftp, smtp, ssh, dns, ftp-data ,irc and (-) if not much used service
Transport protocol	tcp	+0.323	supports	Transaction protocol

LIME: not generated — LIME was only run in research Notebook 16 for an older detector; it is not computed per flow for the deployed model.

D. What the sources explain

Passages were retrieved from MITRE ATT&CK, NIST SP 800-61r3, NIST SP 800-94 and the CISA incident-response playbooks. Statements below were written by the local LLM and passed automated checks against the cited passage (lexical overlap, not proof of meaning). Class-to-technique mappings are hypotheses for context, not proof that a technique was used.

DoS (flow #79)

Mapped techniques (hypothesis): DoS: T1498 Network Denial of Service, T1499 Endpoint Denial of Service

ATT&CK techniques selected by the LLM

• T1498 Network Denial of Service [S1]

What such an attack does

• Adversaries may perform Network Denial of Service (DoS) attacks to degrade or block the availability of targeted resources to users. [S1]

Supporting passage: “Adversaries may perform Network Denial of Service (DoS) attacks to degrade or block the availability of targeted

resources to users.”

Exploits (flow #222)

Mapped techniques (hypothesis): Exploits: T1190 Exploit Public-Facing Application, T1210 Exploitation of Remote Services

ATT&CK techniques selected by the LLM

• T1190 Exploit Public-Facing Application [S6]

What such an attack does

• Adversaries may attempt to exploit a weakness in an Internet-facing host or system to initially access a network. [S6]

Supporting passage: “Adversaries may attempt to exploit a weakness in an Internet-facing host or system to initially access a network.”

Generic (flow #142)

Mapped techniques (hypothesis): Generic: T1600 Weaken Encryption

ATT&CK techniques selected by the LLM

• T1600 Weaken Encryption [S14]

What such an attack does

• Adversaries may compromise a network device’s encryption capability in order to bypass encryption that would otherwise protect data communications. [S14]

Supporting passage: “Adversaries may compromise a network device’s encryption capability in order to bypass encryption that would otherwise protect data communications.”

Fuzzers (flow #161)

Mapped techniques (hypothesis): Fuzzers: T1595.002 Vulnerability Scanning, T1499.004 Application or System Exploitation

ATT&CK techniques selected by the LLM

• T1595.002 Vulnerability Scanning [S16]

What such an attack does

• Adversaries may scan victims for vulnerabilities that can be used during targeting. Vulnerability scans typically check if the configuration of a target host/application (ex: software and version) potentially aligns with the target of a specific exploit the adversary may seek to use. [S16]

Supporting passage: “Vulnerability scans typically check if the configuration of a target host/application (ex: software and version) potentially aligns with the target of a specific exploit the adversary may seek to use.”

E. What to do next

Recommendations are not executed by this application. Fields marked "(proposed)" are suggestions of this application, not prescribed by a cited source. No addresses, ports or timestamps exist in this CSV, so no blocking rules are generated.

Phase	Action and reason	Applies when · owner · disruption · validation	Citations
Verify A0 · batch	Review the 151 flows whose confidence is below 70% before acting on them (filter: confidence Low). Why: Low-confidence predictions are the most likely to be wrong. origin: analyst_proposed	Always. Owner: SOC analyst (proposed) Disruption: None: read-only review (proposed) Validation: Mark each reviewed flow in the checklist.	-
Verify A1 · flow #79 (DoS)	Check flow #79 (tcp/-, state FIN) against packet captures, host logs or sensor alerts from the same capture period to confirm whether it is DoS activity. Why: The model predicts DoS with probability 0.79; its 90% shortlist is DoS. A prediction is not a confirmed incident. origin: analyst_proposed	Always first: nothing should be blocked on the model prediction alone. Owner: SOC analyst (proposed) Disruption: None: read-only check (proposed) Validation: Record the outcome (confirmed / false positive / unclear) in the checklist.	[F79.E1] [F79.E2] [S22]

Phase	Action and reason	Applies when · owner · disruption · validation	Citations
Contain if confirmed A2 · attack type DoS	Filter Network Traffic: When flood volumes exceed the capacity of the network connection being targeted, it is typically necessary to intercept the incoming traffic upstream to filter out the attack traffic from the legitimate traffic. Why: Guidance retrieved for DoS (see the cited passage). origin: llm_synthesis	Only if the check confirms DoS activity; adapt to the affected systems. Owner: Network / security engineering (proposed) Disruption: May block legitimate traffic that uses the same service or port (proposed) Validation: Confirm the targeted traffic stops and normal services still work; keep a rollback of every change (proposed).	[S3]
Verify A3 · flow #222 (Exploits)	Check flow #222 (tcp/ftp-data, state FIN) against packet captures, host logs or sensor alerts from the same capture period to confirm whether it is Exploits activity. Why: The model predicts Exploits with probability 0.99; its 90% shortlist is Exploits. A prediction is not a confirmed incident. origin: analyst_proposed	Always first: nothing should be blocked on the model prediction alone. Owner: SOC analyst (proposed) Disruption: None: read-only check (proposed) Validation: Record the outcome (confirmed / false positive / unclear) in the checklist.	[F222.E1] [F222.E2] [S22]
Contain if confirmed A4 · attack type Exploits	Application Isolation and Sandboxing: Application isolation will limit what other processes and system features the exploited target can access. Why: Guidance retrieved for Exploits (see the cited passage). origin: llm_synthesis	Only if the check confirms Exploits activity; adapt to the affected systems. Owner: Network / security engineering (proposed) Disruption: May block legitimate traffic that uses the same service or port (proposed) Validation: Confirm the targeted traffic stops and normal services still work; keep a rollback of every change (proposed).	[S9]
Contain if confirmed A5 · attack type Exploits	Filter Network Traffic: Restrict outbound network traffic from public-facing servers to prevent unauthorized connections from initiating communications with attacker-controlled infrastructure. Why: Guidance retrieved for Exploits (see the cited passage). origin: llm_synthesis	Only if the check confirms Exploits activity; adapt to the affected systems. Owner: Network / security engineering (proposed) Disruption: May block legitimate traffic that uses the same service or port (proposed) Validation: Confirm the targeted traffic stops and normal services still work; keep a rollback of every change (proposed).	[S11]
Verify A6 · flow #142 (Generic)	Check flow #142 (udp/dns, state INT) against packet captures, host logs or sensor alerts from the same capture period to confirm whether it is Generic activity. Why: The model predicts Generic with probability 1.00; its 90% shortlist is Generic. A prediction is not a confirmed incident. origin: analyst_proposed	Always first: nothing should be blocked on the model prediction alone. Owner: SOC analyst (proposed) Disruption: None: read-only check (proposed) Validation: Record the outcome (confirmed / false positive / unclear) in the checklist.	[F142.E1] [F142.E2] [S22]
Contain if confirmed A7 · attack type Generic	Change system admin passwords, rotate private keys and service/application account secrets where compromise is suspected. Why: Guidance retrieved for Generic (see the cited passage). origin: llm_synthesis	Only if the check confirms Generic activity; adapt to the affected systems. Owner: Network / security engineering (proposed) Disruption: May block legitimate traffic that uses the same service or port (proposed) Validation: Confirm the targeted traffic stops and normal services still work; keep a rollback of every change (proposed).	[S13]

Phase	Action and reason	Applies when · owner · disruption · validation	Citations
Contain if confirmed A8 · attack type Generic	Prevent Domain Name Server (DNS) resolution of known attacker domain names. Why: Guidance retrieved for Generic (see the cited passage). origin: llm_synthesis	Only if the check confirms Generic activity; adapt to the affected systems. Owner: Network / security engineering (proposed) Disruption: May block legitimate traffic that uses the same service or port (proposed) Validation: Confirm the targeted traffic stops and normal services still work; keep a rollback of every change (proposed).	[S13]
Verify A9 · flow #161 (Fuzzers)	Check flow #161 (tcp/-, state FIN) against packet captures, host logs or sensor alerts from the same capture period to confirm whether it is Fuzzers activity. Why: The model predicts Fuzzers with probability 0.96; its 90% shortlist is Fuzzers. A prediction is not a confirmed incident. origin: analyst_proposed	Always first: nothing should be blocked on the model prediction alone. Owner: SOC analyst (proposed) Disruption: None: read-only check (proposed) Validation: Record the outcome (confirmed / false positive / unclear) in the checklist.	[F161.E1] [F161.E2] [S22]
Contain if confirmed A10 · attack type Fuzzers	Efforts should focus on minimizing the amount and sensitivity of data available to external parties. Why: Guidance retrieved for Fuzzers (see the cited passage). origin: llm_synthesis	Only if the check confirms Fuzzers activity; adapt to the affected systems. Owner: Network / security engineering (proposed) Disruption: May block legitimate traffic that uses the same service or port (proposed) Validation: Confirm the targeted traffic stops and normal services still work; keep a rollback of every change (proposed).	[S19]
Recover A11 · batch	20 See NIST SP 800-184: Guide for Cybersecurity Event Recovery for additional guidance. - Developing response scenarios for threat actor use of alternative attack vectors. - Allowing adequate time to ensure all systems are clear of all possible threat actor persistence mechanisms (backdoors, etc.) as adversaries often use more than one mechanism. ... Why: Recovery guidance retrieved for a confirmed incident. origin: retrieved_guidance	Only after a confirmed incident has been contained. Owner: System owners / incident response lead (proposed) Disruption: May need planned downtime (proposed) Validation: Check that restored systems behave normally and that monitoring sees no recurrence (proposed).	[S23]

General prevention (not specific to these flows)

• **attack type DoS:** Filter Network Traffic: When flood volumes exceed the capacity of the network connection being targeted, it is typically necessary to intercept the incoming traffic upstream to filter out the attack traffic from the legitimate traffic. ... [S3]

• **attack type Exploits:** Disable or Remove Feature or Program: Minimize available services to only those that are necessary. [S8]

• **attack type Exploits:** Application Isolation and Sandboxing: Application isolation will limit what other processes and system features the exploited target can access. [S9]

• **attack type Exploits:** Vulnerability Scanning: Regularly scan the internal network for available services to identify new and potentially vulnerable services. [S10]

• **attack type Exploits:** Filter Network Traffic: Restrict outbound network traffic from public-facing servers to prevent unauthorized connections from initiating communications with attacker-controlled infrastructure. ... [S11]

• **attack type Fuzzers:** Filter Network Traffic: Leverage services provided by Content Delivery Networks (CDN) or providers specializing in DoS mitigations to filter traffic upstream from services. Filter boundary traffic by blocking source addresses sourcing the attack, blocking port [S18]

• **attack type Fuzzers:** Pre-compromise: This technique cannot be easily mitigated with preventive controls since it is based on behaviors performed outside of the scope of enterprise defenses and controls. Efforts should focus on minimizing the amount and sensitivity of data availabl [S19]

F. Limits, sources and provenance

What is not known

• No IP addresses, ports, host names, timestamps or payloads are in the uploaded file, so hosts, timelines and botnet or DNS attribution cannot be established.

• Predictions are not confirmed incidents; class-to-ATT&CK mappings are hypotheses for context.

• Heuristic severity and triage priority are application rules, not measured impact.

• Illustrative flows cover 5 of 300 flows; other flows of the same type may differ.

Coverage: 5 of 300 flows have an evidence card; LLM explanations generated for 4 of 9 detected attack types. Human review: not reviewed yet (see the action checklist).

Source documents

Source	Publisher and title	Version / dates	Document link and hash	Cited
CISA IR Playbooks	CISA Cybersecurity Incident & Vulnerability Response Playbooks	- PDF created 2024-08-23 retrieved 2026-10-18	download sha256 2277247542d844d2258c...	6
MITRE ATT&CK	The MITRE Corporation MITRE ATT&CK, Enterprise matrix (STIX 2 bundle)	matrix object modified 2026-07-31; newest technique modified 2026-10-12 retrieved 2026-10-18	download sha256 66f3fe3f07ef1a2de0bd...	7
MITRE ATT&CK mitigation	The MITRE Corporation MITRE ATT&CK, Enterprise matrix (STIX 2 bundle)	matrix object modified 2026-07-31; newest technique modified 2026-10-12 retrieved 2026-10-18	download sha256 66f3fe3f07ef1a2de0bd...	7
NIST SP 800-61r3	NIST Incident Response Recommendations and Considerations for Cybersecurity Risk Management: A CSF 2.0 Community Profile	SP 800-61 Revision 3 PDF created 2025-04-01 retrieved 2026-10-18	download sha256 e5593d6bb85daecec7e8...	1
NIST SP 800-94	NIST NIST SP 800-94, Guide to Intrusion Detection and Prevention Systems (IDPS)	SP 800-94 PDF created 2007-02-15 retrieved 2026-10-18	download sha256 289c2ed27f7e3aac4162...	2

Source register (passages cited in this report)

Id	Source and location	Retrieval	Passage (excerpt; full text in the web view and JSON)
S1	MITRE ATT&CK T1498 sha256 fe2ae28205dada5a...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1498 Network Denial of Service. Tactics: impact. Adversaries may perform Network Denial of Service (DoS) attacks to degrade or block the availability of targeted resources to users. Network DoS can be performed by exhausting the network bandwidth services rely on. Example resources include specific websites, email services, DNS, and web-based applications. ...
S2	MITRE ATT&CK T1499 sha256 12926c3ef5633f0d...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1499 Endpoint Denial of Service. Tactics: impact. Adversaries may perform Endpoint Denial of Service (DoS) attacks to degrade or block the availability of services to users. Endpoint DoS can be performed by exhausting the system resources those services are hosted on or exploiting the system to cause a persistent crash condition. ...
S3	MITRE ATT&CK mitigation M1037 Filter Network Traffic (for T1498) sha256 9a1c430b2caf8c55...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Filter Network Traffic: When flood volumes exceed the capacity of the network connection being targeted, it is typically necessary to intercept the incoming traffic upstream to filter out the attack traffic from the legitimate traffic. ...
S4	NIST SP 800-94 page 59 sha256 3b0e92227cdc5f2c...	TF-IDF similarity to the class and technique names; score 0.1284	They cannot detect the use of passive sensors that simply monitor and analyze observed traffic.38 Denial of service (DoS) attacks and conditions (e.g., network interference). ...

Id	Source and location	Retrieval	Passage (excerpt; full text in the web view and JSON)
S5	NIST SP 800-94 page 68 sha256 b5b479ad633d5f3e...	TF-IDF similarity to the class and technique names; score 0.1194	These attacks typically involve significantly increased bandwidth usage or a much larger number of packets or connections to or from a particular host than usual. By monitoring these characteristics, anomaly detection methods can determine if the observed activity is significantly different than the expected activity. ...
S6	MITRE ATT&CK T1190 sha256 d3a700fe87e6d49c...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1190 Exploit Public-Facing Application. Tactics: initial-access. Adversaries may attempt to exploit a weakness in an Internet-facing host or system to initially access a network. The weakness in the system can be a software bug, a temporary glitch, or a misconfiguration. ...
S7	MITRE ATT&CK T1210 sha256 b79a833daefa4ece...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1210 Exploitation of Remote Services. Tactics: lateral-movement. Adversaries may exploit remote services to gain unauthorized access to internal systems once inside of a network. ...
S8	MITRE ATT&CK mitigation M1042 Disable or Remove Feature or Program (for T1210) sha256 f018d527bce7c9b3...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Disable or Remove Feature or Program: Minimize available services to only those that are necessary.
S9	MITRE ATT&CK mitigation M1048 Application Isolation and Sandboxing (for T1190) sha256 b46ba7e2a134f815...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Application Isolation and Sandboxing: Application isolation will limit what other processes and system features the exploited target can access.
S10	MITRE ATT&CK mitigation M1016 Vulnerability Scanning (for T1210) sha256 a0184902520e34b2...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Vulnerability Scanning: Regularly scan the internal network for available services to identify new and potentially vulnerable services.
S11	MITRE ATT&CK mitigation M1037 Filter Network Traffic (for T1190) sha256 caf776bda7df18de...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Filter Network Traffic: Restrict outbound network traffic from public-facing servers to prevent unauthorized connections from initiating communications with attacker-controlled infrastructure. ...
S12	CISA IR Playbooks page 12 sha256 69a6602c3ec22323...	TF-IDF similarity to the class and technique names; score 0.1603	Table 1: Example Adversary Tactics, Techniques, and Relevant Log and Event Data Tactic Common Techniques Log and Event Sources Indicators Initial Access Phishing [T1566], Drive-by Compromise [T1189], Exploit Public Facing Application [T1190], External Remote Services [T1133] Email, web proxy, server application logs, IDS/IPS Phishing, redirect, and payload ...
S13	CISA IR Playbooks page 31 sha256 027e7cbc3437f766...	TF-IDF similarity to the class and technique names; score 0.1053	TLP:CLEAR Step Incident Response Procedure Action Taken Date Completed 7c. Coordinate with law enforcement to collect and preserve evidence (as required by (Step 3a) prior to eradication, if applicable. 7d. ...
S14	MITRE ATT&CK T1600 sha256 f848c7fb14770438...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1600 Weaken Encryption. Tactics: defense-impairment. Adversaries may compromise a network device’s encryption capability in order to bypass encryption that would otherwise protect data communications. ...
S15	CISA IR Playbooks page 31 sha256 8ea4c248cca62261...	TF-IDF similarity to the class and technique names; score 0.1022	7k. Monitor for signs of threat actor response to containment activities. 7l. Report updated timeline and findings (including new atomic and behavioral indicators) to CISA. 7m. If new signs of compromise are found, return to technical analysis (Step 4) to re-scope the incident. 7n. ...
S16	MITRE ATT&CK T1595.002 sha256 c5c97b4f3b640ef2...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1595.002 Vulnerability Scanning. Tactics: reconnaissance. Adversaries may scan victims for vulnerabilities that can be used during targeting. Vulnerability scans typically check if the configuration of a target host/application (ex: software and version) potentially aligns with the target of a specific exploit the adversary may seek to use. ...
S17	MITRE ATT&CK T1499.004 sha256 dfb90bafb149428b...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1499.004 Application or System Exploitation. Tactics: impact. Adversaries may exploit software vulnerabilities that can cause an application or system to crash and deny availability to users. ...
S18	MITRE ATT&CK mitigation M1037 Filter Network Traffic (for T1499.004) sha256 1e822ce891cbf5f4...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Filter Network Traffic: Leverage services provided by Content Delivery Networks (CDN) or providers specializing in DoS mitigations to filter traffic upstream from services. Filter boundary traffic by blocking source addresses sourcing the attack, blocking ports that are being targeted, or blocking protocols being used for transport.

Id	Source and location	Retrieval	Passage (excerpt; full text in the web view and JSON)
S19	MITRE ATT&CK mitigation M1056 Pre-compromise (for T1595.002) sha256 e00239a46a941968...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Pre-compromise: This technique cannot be easily mitigated with preventive controls since it is based on behaviors performed outside of the scope of enterprise defenses and controls. Efforts should focus on minimizing the amount and sensitivity of data available to external parties.
S20	CISA IR Playbooks page 23 sha256 2883387114bc8170...	TF-IDF similarity to the class and technique names; score 0.1248	The system is not vulnerable. - Susceptible. The system is vulnerable, but no signs of exploitation were found, and remediation has begun. - Compromised. The system was vulnerable, signs of exploitation were found, and incident response and vulnerability remediation has begun. ...
S21	CISA IR Playbooks page 23 sha256 d693aa3b2b7962a4...	TF-IDF similarity to the class and technique names; score 0.099	If the vulnerability exists in the environment, address the vulnerability itself—as described in the Remediation section below—and determine whether it has been exploited in the agency's environment. Use existing best practices to find signs of exploitation, including: - A sweep for known IOCs associated with exploitation of the vulnerability. ...
S22	NIST SP 800-61r3 page 36 sha256 ce7f9c9d56dae692...	TF-IDF similarity to a response-phase query; score 0.1051	R1: Track and validate the status of all ongoing incidents so that incidents in need of more response resources or a change in response strategy can be identified and the necessary changes initiated rapidly. ...
S23	CISA IR Playbooks page 15 sha256 e76b16c25038117a...	TF-IDF similarity to a response-phase query; score 0.186	20 See NIST SP 800-184: Guide for Cybersecurity Event Recovery for additional guidance. - Developing response scenarios for threat actor use of alternative attack vectors. - Allowing adequate time to ensure all systems are clear of all possible threat actor persistence mechanisms (backdoors, etc.) as adversaries often use more than one mechanism. ...

Scraping/ingestion collected these documents; retrieval selected passages; the LLM may only cite them. A retrieval score measures word similarity, not the probability that a claim is true.

Provenance

Item	Value
Detector	LightGBM v3 (class offsets, chosen without test labels) (v3); bundle built 2026-10-26T14:48:53+00:00 from commit be09e74cca
Model files (sha256)	model.txt 8ff9a303ef36d60f...; preprocessing.json de04362be907e9d7...; calibration.json d2a5c2658b69d838...; model_card.json 18177a43efd705fe...
Preprocessing	42 UNSW-NB15 flow features + fe_total_bytes, fe_byte_balance, encoded into 195 model columns
Calibration	temperature 1.159; class offsets [-0.9, -0.9, 0.8, 0.0, -0.2, 0.2, 0.0, -0.4, -0.8, 0.0]; alpha 0.1
Explanation	TreeSHAP (LightGBM pred_contrib) of the base booster, grouped by original column; exact for the deployed class score: (base value + contributions) / temperature + class offset; not a decomposition of the probability
Knowledge base	downloaded 2026-10-18; chunks sha256 73a33392c414a28b...
LLM	qwen2.5:7b (local Ollama), temperature 0, seed 42; uploaded flows are sent only to the local model
Input file	e8f8eb325c36dd3c83e88c0d9a9aebce006d49f6ad67887c873aa7c327e61700
Application	code commit be09e74cca; report schema 2.0
Timings (s)	detections_summary 0.016, xai 0.402, retrieval 0.054, generation 33.743, checks 0.02, total_record 34.283

Appendix 1. Comparison with the labels in the uploaded file

Labels found in the uploaded file were compared with the predictions after the analysis. They were not used for predictions, explanations, retrieval or actions.

Check	Result
Attacks predicted as attacks	261 of 270 (96.7%)
Normal flows predicted as attacks	5 of 30 (16.7%)
Exact class right	177 of 300 (59.0%)
True class on the 90% shortlist	244 of 300

Appendix 2. Model benchmark (context, not measured on this upload)

Benchmark results on the official UNSW-NB15 test file: context for the model, not performance measured on this upload.

Metric	Attack vs normal	10-class macro	10-class weighted
Accuracy	86.5%	74.5%	74.5%
Precision	81.5%	53.8%	84.2%
Recall	97.6%	58.6%	74.5%
F1	88.8%	54.1%	77.9%

Best possible on that file (a model that already knows the labels): accuracy 93.6%, macro F1 76.5%. Top-2 accuracy 91.2%. Normal flows flagged: 27.2%.

Confusion matrix (official test file)

True class (rows) / predicted (columns)	1	2	3	4	5	6	7	8	9	10
1. Analysis	50	71	500	40	0	0	16	0	0	0
2. Backdoor	50	45	462	19	1	0	1	0	5	0
3. DoS	146	867	2,427	516	43	8	20	29	32	1
4. Exploits	185	804	2,620	7,063	110	27	71	169	79	4
5. Fuzzers	112	132	1,059	274	3,333	4	946	5	197	0
6. Generic	0	5	140	354	26	18,317	5	0	19	5
7. Normal	482	1	227	654	8,594	7	26,937	5	93	0
8. Reconnaissance	13	114	231	273	12	1	11	2,808	30	3
9. Shellcode	0	0	21	21	21	0	5	1	308	1
10. Worms	0	0	1	16	0	0	0	0	0	27

Columns use the same numbers as the rows: 1 = Analysis; 2 = Backdoor; 3 = DoS; 4 = Exploits; 5 = Fuzzers; 6 = Generic; 7 = Normal; 8 = Reconnaissance; 9 = Shellcode; 10 = Worms. Green diagonal = correct; red = mistakes (darker = larger share of that true class).